

Make Clawd Dance!

Mode 1 — for directors age 8 and up (with a grown-up helper) · 模式一：给 8 岁以上的小导演（需要大人帮忙）

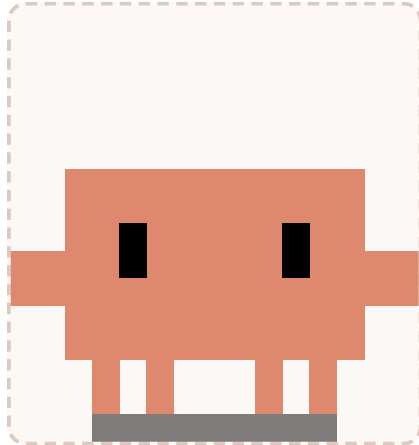
You are going to teach Clawd the crab a brand-new dance. You don't need to know how to code. You are the *director*: you dream up the dance, an AI artist draws it, and you decide if it's good enough. Real directors make the artist try again — a lot!

你将教小螃蟹 Clawd 一支全新的舞！你不需要会写代码。你是**导演**：你来想舞蹈，AI 画家来画，你来决定好不好。真正的导演会让画家改很多次哦！

Meet Clawd 认识 Clawd

Clawd is made of only **rectangles** — like pixel building blocks. He has a body, two claws, four legs, two eyes, and a shadow. When he dances, the rectangles move — they never change shape into circles. That's the rule of his world!

Clawd 全身都是**长方形**，就像像素积木。他有身体、两只钳子、四条腿、两只眼睛和一个影子。跳舞时长方形会动，但永远不会变成圆形——这是他世界的规则！



This is Clawd standing still — his "base pose". 这是 Clawd 站着不动的样子——他的“基本姿势”。

Your artist is an AI 你的画家是 AI

The artist who draws your dance is **Claude** — an AI (artificial intelligence) computer program, not a person. It learned from millions of examples, it can't see pictures (it writes *code* that your browser turns into the dance), it sometimes misunderstands you, and it can give a different answer to the same question. Knowing that is your director superpower: look carefully, say it clearer, try again.

给你画舞蹈的画家是 **Claude**——一个 AI（人工智能）电脑程序，不是真人。它从海量例子中学习；它看不见图画（它写的是代码，浏览器把代码变成舞蹈）；它有时会理解错；同一个问题每次的回答也可能不一样。知道这些就是你的导演超能力：仔细看、说清楚、再试一次。

Want to see finished dances first? 想先看看做好的舞蹈吗？

Ask your grown-up to open the folder `animation-studio/grammars/rect-crab/examples/` and double-click `clawd-dance-groove.svg` or `clawd-dance-party.svg`. They dance right in the browser!

请大人打开文件夹 `animation-studio/grammars/rect-crab/examples/`，双击里面的文件，Clawd 就会在浏览器里跳舞！

1 Dream up your dance 想出你的舞蹈

Fill in the **Idea Card** on the last page (or just answer the four questions out loud while your grown-up writes them down):

1. **Action** — what is Clawd doing?
2. **Body** — how does his body move?
3. **Eyes** — what are his eyes doing?
4. **Effects** — what magic appears around him? (sparkles? confetti? music notes?)

填好最后一页的**创意卡**（或者大声说出四个问题的答案，让大人帮你写下来）：① 动作——Clawd 在做什么？② 身体——身体怎么动？③ 眼睛——眼睛在做什么？④ 特效——他周围出现什么魔法？

2

Give it a name 给它起名字

Pick a short name with small letters and dashes, like `dance-spin`, `super-jump`, or `robot-dance`. The file will be called `clawd-your-name.svg`.

起一个小写字母加“-”的短名字，比如 `dance-spin`（旋转舞）。文件会叫 `clawd-你的名字.svg`。

3

Ask the AI artist 请 AI 画家画画

Ask your grown-up to open the **Studio page** — then you can do everything yourself, right in the browser!

请大人打开**工作室页面**——然后你可以自己在浏览器里完成一切！

The Studio page (easiest way) 工作室页面（最简单）

The grown-up runs one command:

```
cd animation-studio
python3 tools/studio.py
```

A page opens with this same Idea Card on screen. Type your answers, press “ **Make it dance!**” — and after about a minute, Clawd does your dance right on the page. Press “Change it and try again” to boss the artist around some more. Your ideas are saved as spec cards in `output/specs/`, and every dance lands in `output/`.

大人运行一条命令，屏幕上就会出现这张创意卡。填好答案，按下“ **让它跳舞！**”——大约一分钟后，Clawd 就在页面上跳你的舞！按“改一改再来一次”继续当导演。你的创意会存成卡片放在 `output/specs/`，每支舞都在 `output/` 里。

No Studio page? The two manual ways below work too:

打不开工作室页面？下面两种手动方法也可以：

For the grown-up · 给大人看

Way A — copy & paste (no setup):

```
cd animation-studio
python3 tools/build_prompt.py dance-spin "Clawd spins left and right quickly,
claws out wide, eyes squeezed happy, yellow sparkles pop around him"
```

Open the file it creates in `output/`, copy *everything*, paste it into claude.ai, and send. Save the SVG code Claude returns as `output/clawd-dance-spin.svg` (a plain text file with the `.svg` ending).

Way B — one command (Claude CLI installed):

```
python3 tools/claude_animate.py dance-spin "same description here"
```

It saves the finished file into `output/` by itself.

Put the child's four Idea-Card answers into the description — the more their words, the more it's *their* dance.

4 Watch it dance! 看它跳舞！

Double-click your new file. Your browser opens — and Clawd does **your** dance. Watch it loop a few times!

双击你的新文件，浏览器打开——Clawd 跳的就是**你的**舞！多看几遍循环！

5 Make it better 让它更棒

Not perfect? Great — that's directing! First play detective: watch a whole loop and find ONE thing the AI misunderstood. Then write it in the **"notes to the artist"** box on the Studio page and press "Next take with my notes" — your notes go straight to the AI. Every take is kept in your gallery so you can compare them. Director words that work well:

- "Make it **faster!**" / "slower!"
- "**Bigger** jumps!"
- "**More** sparkles!" / "more confetti!"
- "His eyes should be **closed and happy.**"

不完美？太好了——这就是导演的工作！告诉 AI 要改什么，再来一次：“快一点！”“跳高一点！”“更多闪光！”“眼睛要开心地眯起来！”

Tinker Time (拆开玩玩) — optional!

Your dance file is just text — you can open it and change things yourself! Ask your grown-up to open it in a text editor (Notepad works).

舞蹈文件其实就是文字——你可以自己打开修改！请大人用记事本打开它。

⚠️ First: save a copy! 先存一个备份！

Copy the file and rename it (like `clawd-dance-spin-2.svg`) so you can never lose the good one.

先复制文件、改个名字，这样原来的好版本永远不会丢。

Experiment 1 — Speed 速度

Find a number like `1.6s` (that means 1.6 seconds). Make it smaller, like `0.8s` → twice as fast! Save, then double-click the file again.

找到像 `1.6s` 这样的数字（1.6 秒）。改小它，比如 `0.8s` ——快一倍！保存后再双击看看。

Experiment 2 — Colors 颜色

Find a color code like `#60A5FA` (that's blue). Try `#F87171` (red), `#34D399` (green), or `#FBBF24` (yellow). Never change Clawd's own color `#DE886D` — that's his skin!

找到像 `#60A5FA`（蓝色）的颜色代码，换成红色 `#F87171`、绿色 `#34D399` 或黄色 `#FBBF24`。但不要改 `#DE886D` ——那是 Clawd 的皮肤！

Experiment 3 — Bigger moves 更大的动作

Find `rotate(-22deg)` or `translateY(-3.5px)`. Bigger numbers = bigger moves. Try `rotate(-45deg)`. What happens at `rotate(-360deg)`? 😊

找到 `rotate(-22deg)`（旋转）或 `translateY(-3.5px)`（跳高）。数字越大动作越大。试试 `rotate(-360deg)` 会怎样？😊

Science Time: the Clear-Words Experiment 科学时间：清楚话实验

Real scientists test things. Try this experiment on the Studio page:

1. **Vague take:** fill the whole card with just "*Clawd dances.*" Press the button. Look at what you get.

2. **Clear take:** now describe YOUR dance exactly — "Clawd hops on his left legs, claws punching the sky one at a time, eyes squeezed happy, red stars popping around his claws."
3. **Compare** the two takes in your gallery. Which one looks more like what you imagined? Why?

What you discovered: the AI cannot read your mind — it can only read your words. Vague words make the AI guess; clear words put YOU in charge. Scientists call the guessing "filling in with its own patterns." Directors call it "not what I asked for!" 😊

真正的科学家会做实验。在工作室页面试试：① **模糊版**：整张卡片只写“Clawd 跳舞”，按下按钮，看看结果。② **清楚版**：把你的舞蹈准确描述出来——“Clawd 用左边的腿跳，钳子轮流打向天空，眼睛开心地眯着，红色星星在钳子旁边爆开。”③ 在画廊里**比较**两版：哪一版更像你想象的？为什么？

你的发现：AI 不会读心术——它只会读你的字。话说得模糊，AI 就自己猜；话说得清楚，导演就是你。这就是为什么“把话说清楚”是和 AI 合作最重要的本事！

🧠 How did Claude learn to do this? Claude 是怎么学会的？

Claude was not built knowing how to make Clawd dance. It **learned from examples** — it read more words and code than any person could read in a thousand lifetimes. From all that reading, it learned **patterns**.

You do this too! If someone says "Once upon a..." — you already know the next word is "time." You learned that pattern from hearing lots of stories. Claude works the same way, just with many, many more patterns: patterns about words, about code, and even about how dancing crabs should bounce.

Knowing this explains Claude's superpowers AND its silly mistakes:

- It has seen SO many patterns that it can write dance code in seconds. 🚀
- It has no eyes — it has never actually seen Clawd dance! It only predicts what good dance code looks like. 👁️🚫
- When it's unsure, it doesn't say "I don't know" — it makes its best guess. Sometimes the guess is wrong, and that's when the director (you!) catches it. 🤖
- It doesn't remember your last dance. Every Idea Card sends ALL the rules again — peek at the prompt file in `output/` and you'll see the whole rulebook goes along every single time! 📖

Claude 不是生来就会让 Clawd 跳舞的，它是**从例子中学会的**——它读过的文字和代码，比一个人一千辈子能读的还多，并从中学到了**规律**。你也会这样！有人说“很久很久……”，你就知道下一个词是“以前”。你从很多故事里学到了这个规律。Claude 也一样，只是它的规律多得多。知道这一点，就明白了它的超能力和小失误：① 规律见得多，几秒钟就能写出舞蹈代码；② 它没有眼睛——它从没“看见”过 Clawd 跳舞，只是预测好的舞蹈代码长什么样；③ 不确定时它不说“我不知道”，而是给出最好的猜测——猜错的时候就靠导演（你！）来发现；④ 它不记得上一支舞——每张创意卡都会把全部规则重新发一遍，去 `output/` 看看提示词文件，整本规则书每次都跟着走！

Director's checklist 导演检查表

- ☐ It dances when I double-click it. 双击就会跳舞。
- ☐ The dance repeats smoothly, no sudden jump. 循环很顺滑，没有突然跳一下。
- ☐ It still looks like Clawd. 看起来还是 Clawd。
- ☐ Nothing falls off the edge of the picture. 没有东西跑出画面。
- ☐ I'm proud of it! 我很自豪！



Idea Card 创意卡

Name 名字: clawd-_____.svg

1. Action — What is Clawd doing? 动作——Clawd 在做什么？

.....

2. Body — How does his body move? 身体——身体怎么动？

.....

3. Eyes — What do his eyes do? 眼睛——眼睛在做什么？

.....

4. Effects — What appears around him? 特效——周围出现什么？

.....

Speed 速度: ☐ slow 慢 ☐ medium 中 ☐ FAST! 快！

Mood 心情: ☐ happy 开心 ☐ silly 搞怪 ☐ sleepy 想睡 ☐ excited 激动